Hours of Account

Watchmakers, Ledgers and the Moral Time of Geneva, 1760–1790

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> *Wound the great clock at six, and my own at a quarter past, so that the house should not run ahead of the city.*
>
> — Abraham Courvoisier, *Livre de raison*, 14 March 1771

Historians of time have long told a story about discipline. In E. P. Thompson's classic account, the spread of clocks and watches in the eighteenth century accompanied a new regime of work in which the employer's hour displaced the rhythms of task and season, and the moral language of punctuality was the language of the factory.[1] The story has been qualified many times, most usefully by work showing that clock time was familiar to ordinary people long before the mills, and that it served them as often as it bound them.[2] Yet the qualification tends to keep the original question: who imposed the hour on whom?

This essay asks a different question of a different place. Geneva in the second half of the eighteenth century was the watchmaking capital of Europe, a republic whose workshops, the *Fabrique*, exported tens of thousands of movements a year.[3] Its master watchmakers did not need to be taught the hour; they made it. What they kept, besides their clocks, were account books, and the account books show something that neither the factory nor the parish bell explains: an attempt to make time itself into a moral account, kept in the same columns as debts and obligations.

1. E. P. Thompson, "Time, Work-Discipline, and Industrial Capitalism," *Past & Present*, no. 38 (1967): 56–97.

2. Paul Glennie and Nigel Thrift, *Shaping the Day: A History of Timekeeping in England and Wales, 1300–1800* (Oxford: Oxford University Press, 2009), 1–24.

3. David S. Landes, *Revolution in Time: Clocks and the Making of the Modern World* (Cambridge, MA: Harvard University Press, 1983), 237–245.

I

THE LEDGER AS A CLOCK

The richest of these books is the *livre de raison* of Abraham Courvoisier, a master in the quarter of Saint-Gervais who kept it from 1764 until his death in 1789.[4] It is a domestic ledger of a familiar kind, recording rents, wages, the dowry of a daughter and the price of firewood. What makes it unusual is that roughly one entry in five records not money but time: hours worked by journeymen, hours lost to illness, hours "given" to the consistory or the militia, and the daily setting of the household clock against the bell of the cathedral of Saint-Pierre.

Courvoisier entered these hours in the same hand, and in the same double columns, as his debts. Time owed to the workshop sits beside money owed to the supplier of brass, and both are struck through when the debt is settled. The ledger thus treats an hour as something that can be lent, borrowed and repaid. A characteristic entry for the winter of 1771 reads:

> Pierre Dufour, journeyman, three hours this Tuesday for his mother's burial, which he will make good on Saturday after the bell. Given to the widow Perrin two hours of the boy for her chimney, which she will not repay and need not.[5]

The distinction between hours that "will be made good" and hours that "need not" be repaid is the hinge of the whole book. It separates the economy of the workshop, in which time is a debt like any other, from an economy of charity, in which time is a gift and the gift is recorded precisely so that it can be seen not to be owed.

Readers of Marthe Vuilleumier's study of the *Fabrique* will recognise the structure. Vuilleumier showed that the credit relations between masters, journeymen and the cabinotiers who finished cases were as much a matter of reputation as of money, and that a master's standing depended on being seen to honour obligations he could have evaded.[6] What the Courvoisier ledger adds is that time entered the same economy of reputation. To keep exact account of hours given away was to be known as a man who gave them.

4. Abraham Courvoisier, "Livre de raison, 1764–1789" (Manuscript, Archives d'État de Genève, Commerce F 112, 1771).

5. Courvoisier, "Livre de raison, 1764–1789," fol. 38v.

6. Marthe Vuilleumier, *Les horlogers de la Fabrique: travail, crédit et réputation à Genève au XVIIIe siècle* (Geneva: Éditions du Rhône, 1998), 88–112.

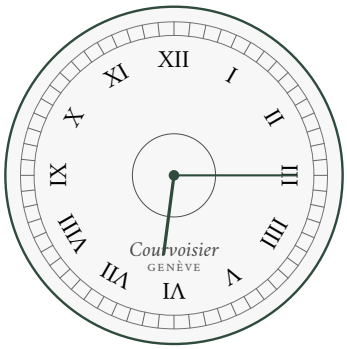


FIGURE 1. Enamel dial of a verge watch signed "Courvoisier, Genève", c. 1775, redrawn from the original. The hands are set to a quarter past six, the time at which the ledger records the winding of the household clock each morning. The four-stroke numeral for four, rather than IV, follows the convention of Genevan dial painters, who balanced the heavier VIII on the opposite side; the signature is placed below the centre so that it remained legible under the hour hand for most of the day.

II

THE REPUBLIC OF THE HOUR

If Courvoisier's ledger turned time into a moral account, it did so within a civic order that already used the hour as a sign of virtue. Geneva's Calvinist institutions had long regulated the day by bell: the hours of sermons, the closing of the city gates, the times at which taverns had to shut.[7] Daniel Okonkwo has argued that by the mid-eighteenth century this regulation had been absorbed into a republican self-image, in which the punctual citizen was the virtuous one, and in which the city's watches, sold across Europe, carried that image abroad.[8]

The ledger complicates this picture in an instructive way. Courvoisier does not set his clock by his own watches, which he made and could have trusted; he sets it by the cathedral bell, and records the difference. Over twenty-five years the entries show the household clock running a few minutes fast in summer and slow in winter, and each correction is noted with the same care as a payment. The household, as the epigraph puts it, "should not run ahead of the city".

Figure 1 shows the dial of one of the watches that left Courvoisier's workshop in these years. Its hands are set at the hour of the morning winding. It is tempting to read the dial as an emblem of private time, carried in a pocket and answering

7. Gerhard Dohrn-van Rossum, *History of the Hour: Clocks and Modern Temporal Orders*, trans. Thomas Dunlap (Chicago: University of Chicago Press, 1996), 201–219.

8. Daniel Okonkwo, "The Punctual Republic: Bells, Watches and Civic Virtue in Calvinist Geneva," in *Cultures of Time in Early Modern Europe*, ed. Lucy Brennan and Hélène Faure (Manchester: Manchester University Press, 2019), 150–156.

to no one. The ledger suggests the opposite reading: the watch was an instrument for keeping one's private hours in step with a public order, and the account book was the record that one had done so.

This is not the discipline Thompson described. Nobody imposed the cathedral's hour on Courvoisier; he imposed it on himself, and he wrote down the result. Imogen Hartwell has called this kind of record "a ledger of conscience", noting that artisanal account books in several Protestant cities recorded hours alongside sums in the decades before industrial work rules existed.[9] The Genevan case shows how close this practice stood to the city's religious and political institutions, and how much it depended on the moral standing of the master who kept the book.

III

CONCLUSION

The history of time-discipline has been written mostly from the point of view of those on whom the clock was imposed. Account books like Courvoisier's allow it to be written from the point of view of those who kept it. What they show is not an early form of factory discipline but a moral economy in which hours, like debts, were owed, repaid and forgiven, and in which the public hour of the city was the standard against which a household measured its own uprightness.

Two consequences follow. First, the precision of Genevan watches, so often treated as a purely technical achievement, belonged to a culture that valued exact accounting of every kind; the watch and the ledger were tools of the same habit of mind.[10] Second, the transition Thompson described, from task-oriented to timed labour, looks less like a rupture when seen from Saint-Gervais. The language of owed and repaid hours was already there; what the factory changed was who kept the account.[11]

9. Imogen Hartwell, "Keeping Accounts, Keeping Hours: Ledgers and the Moral Economy of Artisanal Time," *Journal of Social History* 48, no. 2 (2014): 309–34.

10. Landes, *Revolution in Time: Clocks and the Making of the Modern World*, 245.

11. Thompson, "Time, Work-Discipline, and Industrial Capitalism," 63–70.

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